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MAST20030: Differential Equations
Tutorial: Week 3

Question 1

Consider the ODE

$$x''(t) - x'(t) - 2x(t) = 0.$$

(a) Rewrite this ODE as a vector-valued ODE, i.e., $\boldsymbol{y}'(t) = \boldsymbol{A}\boldsymbol{y}(t)$ where $\boldsymbol{y}(t) = \begin{pmatrix} x(t) \\ v(t) \end{pmatrix}$ where $v(t) = x'(t)$.

(b) Trial a solution of the form $\boldsymbol{y}(t) = \boldsymbol{v}\mathrm{e}^{\lambda t}$ and hence show that λ and $\boldsymbol{v}$ are the eigenvalues and eigenvectors of $\boldsymbol{A}$ respectively.

(c) Calculate these eigenvalues and eigenvectors and hence determine the general solution of the vector-valued ODE. Keep these on the board as you will need them in question 2.

(d) We will now draw the phase portrait to visualise the structure of the solution space of the ODE. Begin by drawing the axes for the $x - v$ plane and then complete the following tasks.

i. Sketch the straight line orbits, i.e., the spans of each eigenvector. What is the direction of the solution along these trajectories? Hint: if a phase point, $\boldsymbol{y}(t)$, is in the span of one of the eigenvectors (i.e., it lies on one of the straight line orbits) then the rate of change vector is given by $\dfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{y}(t) = \boldsymbol{A}\boldsymbol{y}(t) = \lambda\boldsymbol{y}(t)$.

ii. Consider a phase point, $\boldsymbol{y}(t)$, that is not in the span of either of the eigenvectors. Does the linear transformation, $\boldsymbol{A}$, rotate the phase point? Hence do you expect the trajectories away from the eigenvectors to be straight lines or curves?

iii. Again, consider a phase point, $\boldsymbol{y}(t)$, that is not in the span of either of the eigenvectors. What happens to the trajectory that this phase point lies on in the limits $t \to -\infty$ and $t \to \infty$? Hint: examine the general solution in these two limits.

iv. Using the insight gained from ii and iii sketch some trajectories that do not lie along the span of either of the eigenvectors, thereby completing the phase portrait.

v. Have a brief discussion with your group on how linear algebra dictates the structure of the solution space of this ODE.

Question 2

We now consider the related inhomogeneous ODE

$$x''(t) - x'(t) - 2x(t) = \mathrm{e}^{-t}$$

(a) Rewrite this ODE as a vector-valued inhomogeneous ODE.

(b) Diagonalise $\boldsymbol{A}$ using the eigenvalues and eigenvectors you found in question 1, i.e., $\boldsymbol{A} = \boldsymbol{S}\boldsymbol{D}\boldsymbol{S}^{-1}$.

(c) By introducing a new vector-valued quantity, $\boldsymbol{z} = \boldsymbol{S}^{-1}\boldsymbol{y}$, rewrite the vector-valued ODE as a set of two decoupled ODEs.

(d) Solve these decoupled ODEs and thereby determine the solution of the vector-valued ODE. You may leave your final answer as a product of matrices.

Question 3

In this question we will consider a harmonic oscillator with no damping

$$x''(t) + 4x(t) = 0$$

(a) Rewrite this ODE as a vector-valued ODE.

(b) By trialling a solution of the form $\boldsymbol{y}(t) = \boldsymbol{v}\mathrm{e}^{\lambda t}$, find the general solution to the vector-valued ODE. You may leave your answer in terms of complex variables.

(c) Examine the quantities x^2 and v^2. Use these two quantities to determine an implicit equation relating x and v that does not depend on t.

(d) Using your answer from (c) sketch the phase portrait for the ODE. How would you determine the direction of the trajectories? Hint: examine how the linear transformation affects the rate of change vector.

Question 4

In this question we will establish some important properties of the matrix exponential.

(a) Assuming that $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$, show that the matrix exponential satisfies $\mathrm{e}^{\mathbf{A}} = \mathbf{S}\mathrm{e}^{\mathbf{D}}\mathbf{S}^{-1}$.

(b) Evaluate the matrix exponential $\mathrm{e}^{\mathbf{D}}$ for a diagonal matrix $\mathbf{D} = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$.

Question 5

Consider the system

$$\mathbf{r}'(t) + \mathbf{A}\mathbf{r}(t) = \mathbf{0}\,, \quad \mathbf{r}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}, \quad \mathbf{r}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{1}$$

(a) Find the eigenvalues and eigenvectors of $\mathbf{A}$.

(b) Use your answer to (a) and the matrix exponential to solve (1). See also Question 4.

(c) Solve the system (1) again, but this time with $\mathbf{r}(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.